



Advanced Martian construction: High-strength ISRU bricks, robust sulfur bonding, modular assembly, and FEA-verified pyramid habitats

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ARTICLE INFO

Keywords:

Martian construction
Soil simulant
Compaction
Hydration
Pyramid-shaped structures
Sulfur-based bond
FE model

ABSTRACT

Sustainable habitat construction on Mars faces significant challenges, including low atmospheric pressure hindering hydration, reduced gravity complicating compaction, and large habitat pressure differentials. This study presents an integrated In-Situ Resource Utilization (ISRU) approach combining high-strength regolith bricks, hydration-free sulfur bonding, and a modular pyramid habitat design validated by Finite Element Analysis (FEA). Optimized mechanical compaction (40 MPa) of nano-SiO₂-enhanced Martian regolith simulant effectively bypasses hydration constraints, achieving compressive strengths exceeding 20 MPa even at ambient temperatures. A systematic parameter study (pressure, particle size, water content, temperature) yielded predictive design equations and demonstrated potential strength enhancement up to 44.5 MPa with thermal treatment (1000 °C). Furthermore, a robust, hydration-free sulfur-based mortar was developed for modular assembly; optimized flat-cut interfaces yielded bond strengths exceeding 2.0 MPa, crucially shifting the failure mode from the bond interface to the brick material itself (ensuring a reliable minimum tensile capacity >1.2 MPa). Leveraging these advancements, a pyramid-shaped habitat module, advantageous for Martian environmental loads (including a 101.3 kPa internal pressure differential and 3.71 m/s² gravity), was designed. FEA, incorporating experimentally derived material properties (e.g., 22 MPa compressive strength, 1.2 MPa tensile/bond capacity), confirmed the structural integrity, with maximum predicted tensile stress (1.15 MPa) remaining below the bond limit. This research provides a comprehensive, experimentally validated framework—from material development and bonding to structural application—for constructing resource-efficient, durable habitats on Mars, significantly advancing solutions for sustainable extraterrestrial infrastructure.

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1. Introduction

The establishment of sustainable human habitats on Mars has transitioned from conceptual exploration to active development, spurred by ongoing robotic missions and ambitions for crewed exploration [1,2]. Realizing this goal necessitates overcoming significant construction challenges unique to the Martian environment, demanding innovative solutions beyond terrestrial practices [3]. Mars presents formidable obstacles: an extremely tenuous atmosphere (~ 636 Pa) [4], drastic temperature fluctuations (-143 °C– 20 °C) [5–7], and a water phase diagram (Fig. 1 [8]) that precludes stable liquid water on the surface under ambient conditions. This severely hinders conventional construction materials reliant on hydration reactions for strength development. Furthermore, structures must maintain a pressurized internal atmosphere (~ 101 kPa) against the near-vacuum exterior, inducing substantial tensile stresses (internal pressure differential or ‘negative pressure’) in the building envelope. The gravitational pull on Mars measures approximately 3.71 m/s², a stark contrast to Earth’s 9.81 m/s² [8]. This diminished gravity presents challenges in compacting materials, thereby complicating the process of attaining the requisite strength. Moreover, structural designs on Mars need to account for the distinctive attributes of wind, seismic, and gravitational forces. The high-speed winds on Mars, particularly during dust storms, generate considerable dynamic pressures, thus demanding aerodynamic structural configurations to mitigate potential damage [9,10]. Although Martian seismic activity is less intense than that on Earth [11], structures still need to possess flexibility and resilience to absorb and dissipate seismic energy. Therefore, a comprehensive design approach is essential to develop habitats and infrastructure that can withstand and adapt to the unique conditions of Mars [12,13].

One viable approach to constructing durable structures on Mars is to employ 3D printing technology [14] (illustrated in Fig. 2(a)). This approach employs In-Situ Resource Utilization (ISRU) materials, such as Martian regolith, which is rich in silicon dioxide (SiO₂), with concentrations varying between 38 % and 62 % [15–17]. Consolidating regolith typically requires binders. The typical binders includes plaster of Paris (PoP), ordinary Portland cement (OPC), alkali-activated cement (AAC), geopolymers (GC), etc [6,18]. However, these binders typically require liquid water to achieve the necessary compressive strengths. On Mars, where water is present as vapor and ice, liquid brines are only theoretically accessible in limited areas [19,20]. Consequently, water-free binders like elemental sulfur (ES) emerge as more viable alternatives [21–23]. Sulfur concrete, produced by heating abundant Martian sulfides to liquefy sulfur for binding extraterrestrial aggregates, can achieve compressive strengths of up to 35 MPa [24–26]. Optimized mixtures have been shown to exceed 50 MPa in compressive strength [22,27–29]. Despite these strengths, elemental sulfur is prone to freeze-thaw damage [25,30,31] and requires warmer temperatures to prevent sublimation, which can reduce compressive strength by up to 80 % [25,30,31]. Furthermore, sulfate products can cause irritation to the eyes, skin, and mucus membranes upon exposure [32].

An alternative strategy involves pre-fabricating structural elements within a pressurized and temperature-controlled environment. Subsequently, the elements can be transported and assembled at the desired locations on Mars [6]. In this approach, when nano-particulate iron oxides and oxyhydroxides, characterized by a large specific surface area, are compacted and brought into contact, the desired bonding can be achieved [33–35]. Additionally, the process of compacting simulant materials has been shown to yield notable strengths. The compaction process serves multiple crucial functions. Firstly, it is expected that the applied pressure can condense vapor into liquid water. This liquid water is essential as it enables the proper hydration of mixtures, delivering durable constructional elements. Upon compaction of aggregate mixtures that contain liquid water, the aggregate particles become reoriented into a more compact configuration and are forced into closer proximity. This leads to the development of stronger binding forces, which in turn result in the creation of sturdy structural elements. The standardized elements can be manufactured in a factory setting and then

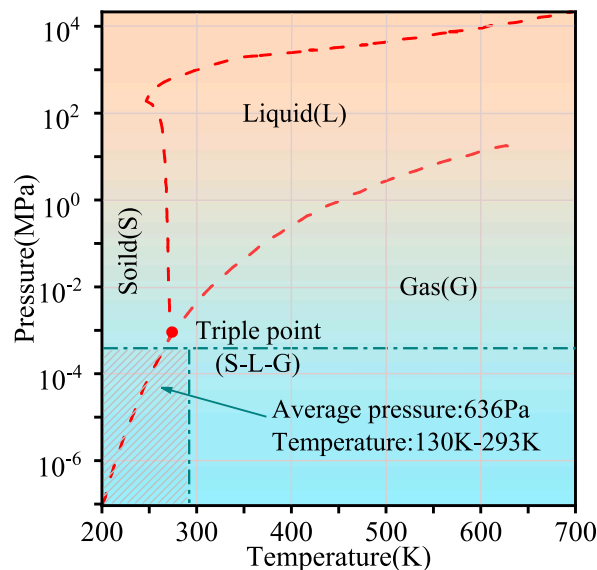
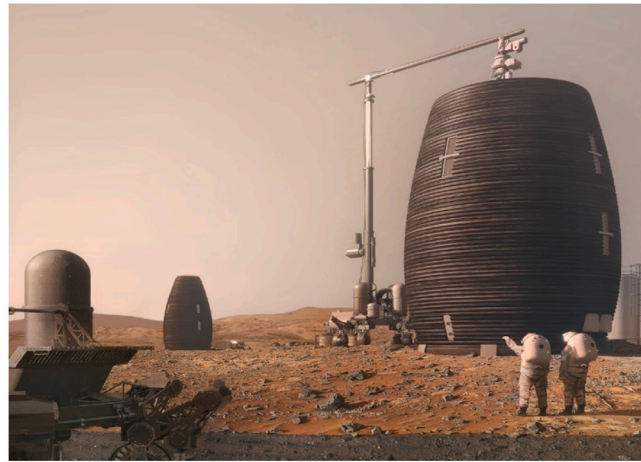


Fig. 1. The pressure-temperature phase diagram of water [8].



(a) Promising 3D printing method by AI Space Factory (<https://www.archdaily.com/910764/ai-spacefactory-builds-3d-printed-mars-prototype-for-nasa>)



(b) Promising 3D printing technologies in use of Martian bricks

Fig. 2. Construction methods.

transported to the field for on-site curing and construction. This approach further simplifies the construction workflow, as illustrated in Fig. 2(b). Special configurations can be prepared to achieve the desired interlocking mechanism for connecting bricks. This not only minimizes the need for mortar binding but also streamlines the printing process, indicating a more rapid construction approach that is better adapted to the Martian environment [36]. Another option is to melt minimal amounts of sulfur to serve as a “Mortar” for bonding Mars bricks [37,38]. This approach requires less sulfur consumption compared to sulfur concrete technology, leading to reduced energy requirements for melting. Furthermore, this methodology offers improved resilience to freeze-thaw cycles and damage caused by sublimation. Notably, in contrast to the dominant 3D printing techniques employed on Mars, this study advocates for the separation of the fabrication of structural elements from the 3D printing process, as illustrated in Fig. 2(b). This decoupling has the potential to enhance operational efficiency and facilitate maintenance for each individual process. However, to date, few studies have delved into exploring the advantages and challenges of this approach. In this context, the proposed bricks are expected to be highly compatible with the Martian atmosphere and exhibit robustness comparable to their terrestrial counterparts. Masonry structures can be readily constructed by leveraging these bricks, along with interlock mechanisms or sulfur-based bond, and 3D printing technology. The proposed methodology, which involves the separation of the manufacturing and construction phases, is expected to enhance the speed of construction while simultaneously reducing maintenance needs. As a result, further investigation in this area will provide a solid foundation for the advancement of a more sustainable, competitive, and efficient construction technology, particularly concerning time and energy utilization on Mars.

In the preliminary stage of design exploration, pyramid-shaped structures can be identified as an outstanding architectural option due to their remarkable resistance to a variety of natural and extraterrestrial forces. One of the key advantages of pyramid-shaped structures lies in their aerodynamic form. This shape enables them to endure high wind loads with great efficiency. By effectively minimizing pressure differentials, the risk of structural damage caused by strong winds is significantly reduced [9,10]. When exposed to the powerful winds prevalent on Mars, especially during dust storms, the aerodynamic design of pyramids helps them maintain stability. The wide base and converging apex of a pyramid play a crucial role in evenly distributing stress. During earthquakes, this design feature reduces torsional effects, allowing the structure to better withstand the dynamic forces generated by seismic activity [39]. In addition to wind and seismic resistance, the dense and solid construction of pyramids offers substantial protection against

cosmic rays and solar wind. As a result, the interior spaces of the pyramid are safeguarded from the harmful effects of radiation, providing a safer environment for potential inhabitants [40]. Pyramid structures are also expected to effectively distribute internal pressure, which implies potential benefits for managing negative pressure. Despite this expectation, there has been a paucity of research to empirically confirm the advantages of pyramid structures in dealing with negative pressure. In this study, Finite element (FE) simulation will be applied to facilitate the design of pyramid-shaped structures. Through the simulation of the Martian gravitational force, which is approximately 3.71 m/s^2 , as well as the combined stress resulting from negative pressure and gravity, it is possible to predict the performance of the structure during both the construction and operational phases. The simulation offers in-depth information regarding the distribution of stress and the potential locations of failure. Leveraging FE simulation, adjustments can be made to the dimensions, material characteristics, and construction methodologies of the pyramids. Such refinements serve to improve the structures' robustness and sustainability within the distinctive Martian milieu.

In summary, this study integrates four critical advancements—novel ISRU material development, optimized sulfur-based bonding, modular assembly strategies, and FE validation—to establish a comprehensive, resource-efficient framework for constructing durable Martian habitats. First, an innovative compaction technique is presented that overcomes Martian environmental constraints (low pressure, low gravity) to produce high-strength ISRU bricks. Subsequent investigations systematically quantify the influence of key manufacturing parameters (pressure, temperature, particle size, water content), yielding predictive design equations. An effective, hydration-free sulfur mortar bonding technique is then characterized and optimized, enabling the robust modular assembly of proposed pyramid-shaped habitats. Finally, FE analysis, critically incorporating experimentally derived material properties and bond strengths, validates the structural viability of the pyramid design under relevant Martian conditions, including gravity and internal pressure differentials. Together, these integrated innovations form a cohesive foundation for advancing sustainable extraterrestrial construction.

2. Experiments

The fabrication of Martian bricks entailed the utilization of composite materials, as detailed in Table 1, which were subjected to a range of pressures, temperatures, hydration levels, particle sizes, and curing conditions. Consistent with the five widely acknowledged simulants of Martian soil[41–45], raw materials with analogous molecular compositions and proportions were combined using a small-scale mixer. Initially, the dry ingredients were agitated for 2 min, followed by the incorporation of water (or fine ice particle on Mars) and further mixing for an additional 3 min to achieve the desired mixtures. All mixtures underwent curing at ambient temperatures of approximately $20\text{ }^{\circ}\text{C}$, a condition that closely mirrors the upper limits of temperatures recorded on the Martian surface [5–7]. A standard soil simulant was chosen to investigate the influence of curing duration. During the compaction testing phase, a consistent pressure was applied for a duration of one and half hours, following the procedures shown in Fig. 3. For specimens subjected to heat treatment, the compacted samples were incrementally heated from ambient temperature to a predetermined target temperature. This temperature was sustained for a prolonged duration of 6 h within an electrically regulated constant temperature drying oven. Following this heating period, the specimens were allowed to cool to ambient room temperature at a controlled, natural rate. In addition to these primary procedures, supplementary experiments were conducted to examine the effects of variations in particle size and water content on the material properties.

Fig. 3 also illustrates the test specimens, which were cubic in shape with dimensions of 25.4 mm (1 inch) on each side. The specimens were crafted utilizing steel molds, which were subsequently filled with Martian soil simulants. The interior cavity of the mold was partitioned into sections measuring 25.4 mm. Additionally, two plates were employed at the extremities to support the loads that were applied. A testing apparatus, capable of exerting a maximum force of 100 tonnes, was utilized to impose the loads. In order to maintain the desired loading conditions, steel reinforcement bars and high-strength bolts were meticulously positioned as shown in Fig. 3. It is expected that solar concentrators will be employed to generate the requisite temperature for the curing process of the structural elements on Mars (see Fig. 2 (b)). These concentrators, known for their cost-effectiveness, have been extensively utilized in various space endeavors, such as the Perseverance rover mission. Subsequently, structural components characterized by advantageous geometries and adequate strength will be conveyed to the construction site for incorporation into 3D printing processes. This approach holds promise for diminishing the volume of materials that need to be transported to Mars, streamlining the construction process, and

Table 1
Martian soil simulants.

	JSC-1m	MGS-1m	JEZ-1m	JMSS-1m	MMS-1m
SiO ₂	46.50	50.30	47.70	52.92	53.26
CaO	6.20	7.40	4.60	7.56	10.45
MgO	3.40	14.60	25.60	6.35	6.08
Fe ₂ O ₃	15.60	–	–	16.00	10.87
FeO _T	–	11.20	11.90	–	–
Al ₂ O ₃	23.30	12.80	8.00	13.64	17.10
P ₂ O ₅	0.90	0.10	0.10	0.30	0.17
TiO ₂	3.80	0.60	0.40	1.78	1.09
MnO	0.30	0.10	0.10	0.14	0.17
Cr ₂ O ₃	–	–	–	0.30	0.05

* all specimens were slightly modified from comparable simulants, e.g. JSC-1, MGS-1, JEZ-1, JMSS-1, and MMS-1 [42–46].

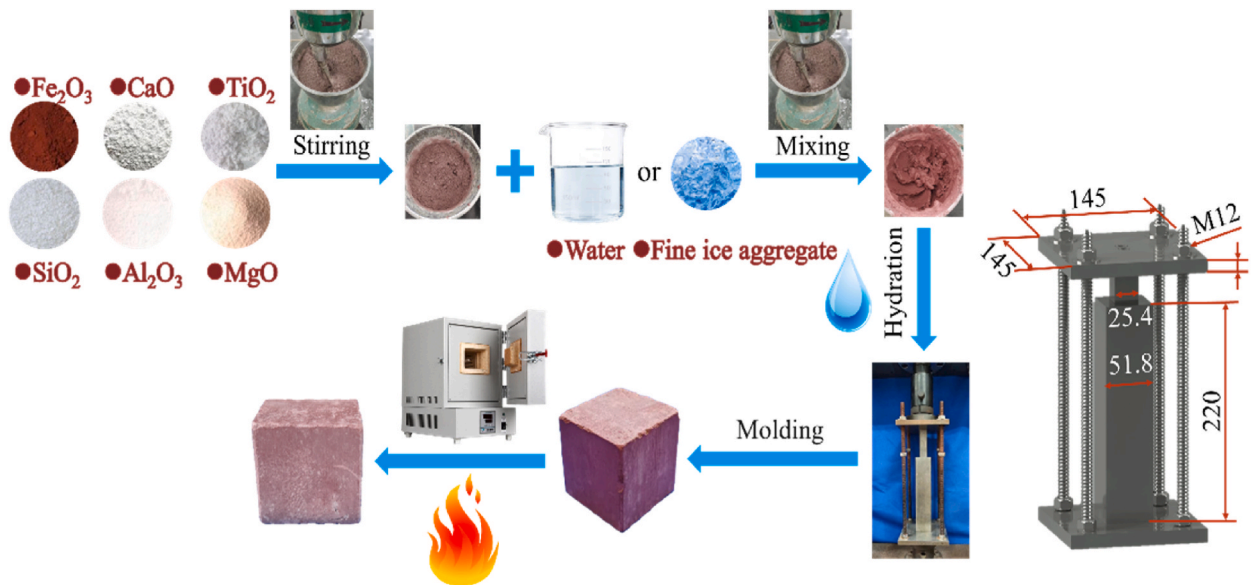


Fig. 3. Construction on Mars including manufacturing apparatus and the resultant specimen from Martian simulants.

reducing the maintenance demands associated with both the manufacturing and construction phases.

Fig. 4 (a) presents the testing methodology employed in this study, where a displacement-controlled procedure was implemented to impose a uniform loading regime on the test specimens at a rate of 0.5 mm/min. The magnitudes of the applied loads were meticulously recorded by the internal load cell integrated within the testing apparatus. Concurrently, the strain experienced by the specimens was recorded by a digital image correlation (DIC) system monitoring the relative displacements between the upper and lower substrate.

In pursuit of an effective bonding mechanism suitable for Martian construction applications, this study also developed a sulfur-based adhesive. Fig. 4 (b) shows the setup for the four-point bending tests conducted to determine the bond strength. Three distinct interface configurations were fabricated to improve the interfacial bond. The interface delivered three types of bonds: a cut-free bond, a flat-cut bond, and a cross-cut bond, all characterized by a bond interface and a cut width measuring 3 mm. To achieve the requisite bending moment for the tests, two cubic bricks were securely attached to steel blocks using epoxy resin. Subsequently, these composite units were bonded together with molten sulfur, adhering to the specified interface conditions. In situ, the sulfur-based bonding agent can be prefabricated to the desired shape and then slightly melted to adapt into the pre-existing bonding interface, thus achieving the desired connection. The experimental setup featured a span of 34 mm between the two loading points, with a support-to-load-point distance of 33 mm. A displacement rate of 0.06 mm/min was meticulously applied to the specimens during the testing phase. Throughout the testing process, the load and displacement data were diligently captured by the internal load cell and the LVDT system, respectively, ensuring a comprehensive dataset for subsequent analysis. This rigorous testing protocol was designed to yield insights into the performance of the bonding technique under Martian construction conditions, thereby informing the development of reliable and robust construction methodologies for extraterrestrial applications.

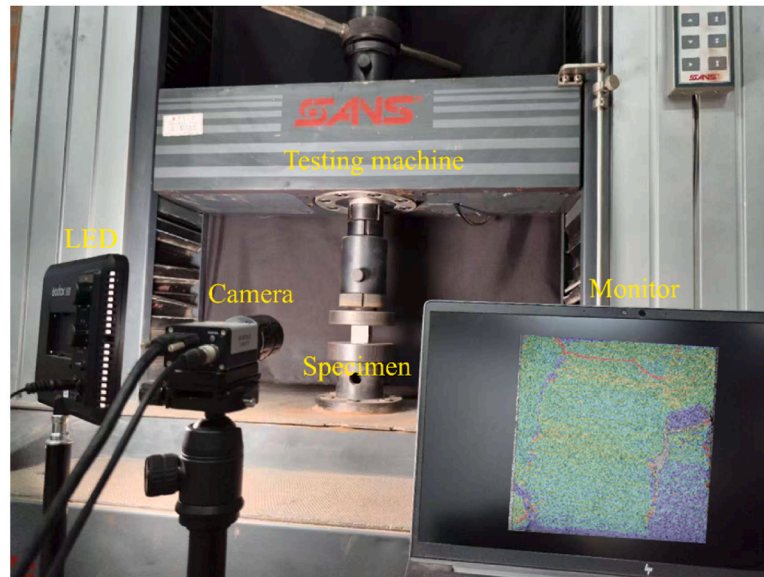
3. Results and discussion

The subsequent section presents the findings obtained from both compressive and flexural testing protocols. The principal objective of these investigations was to develop novel construction materials and adhesive agents suitable for application in the arid, low-pressure, and resource-constrained environment of Mars.

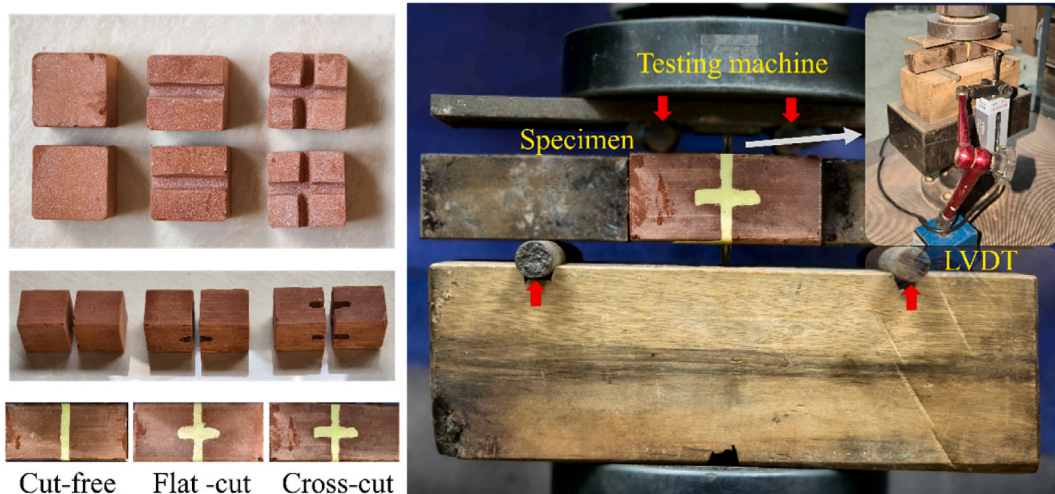
3.1. Simulants

To evaluate the potential of Martian regolith for construction applications and establish baseline properties prior to processing enhancements, five distinct simulant compositions were developed. These formulations, detailed in Table 1, were based on known Martian soil chemistries [42–46], incorporating varied proportions of key oxides. Major components included SiO₂ (ranging from 46.50 % to 53.26 %), Al₂O₃ (8.00 %–23.30 %), FeOT (10.87 %–16.00 %), MgO (3.40 %–25.60 %), and CaO (6.20 %–10.45 %), alongside minor constituents like P₂O₅ and TiO₂. Crucially, initial baseline specimens were prepared at ambient laboratory conditions (~20 °C, 1 atm) without the application of the compaction or thermal treatments investigated later in this study, serving as reference points.

Compressive testing results for these unprocessed baseline specimens are presented in Fig. 5(a) and Table 2. The obtained strengths were consistently low, underscoring the necessity for processing methods like compaction and thermal treatment to achieve structural



(a) Compressive tests



(b) Bond tests

Fig. 4. Testing setup.

viability. For instance, the JSC-1m simulant, possessing the highest Al_2O_3 content (23.3 %), exhibited the highest average baseline compressive strength at 1.7 MPa. Conversely, JEZ-1m, with the lowest Al_2O_3 content, demonstrated the lowest average strength at approximately 1.0 MPa. This variation suggests some influence of chemical composition, particularly Al_2O_3 known for contributing to material strength, but primarily highlights that none of the simulants achieve significant mechanical strength solely through mixing and ambient curing.

Despite the low strength values, the mechanical behavior under compression provided relevant insights. The observed stress-strain characteristics (Fig. 5(b)) typically showed an initial linear-elastic region followed by non-linear behavior leading to the peak stress. Specimens with slightly higher strength also tended to exhibit greater stiffness (steeper initial slope). Failure consistently occurred through brittle crushing (Fig. 5(c)). This overall stress-strain response and failure mode are qualitatively similar to those characteristic of conventional cementitious concrete. While validating the fundamental mechanical nature of the consolidated simulants, these baseline results strongly reinforce the requirement for the advanced processing techniques investigated in subsequent sections to significantly enhance the material's load-bearing capacity for practical construction applications on Mars.

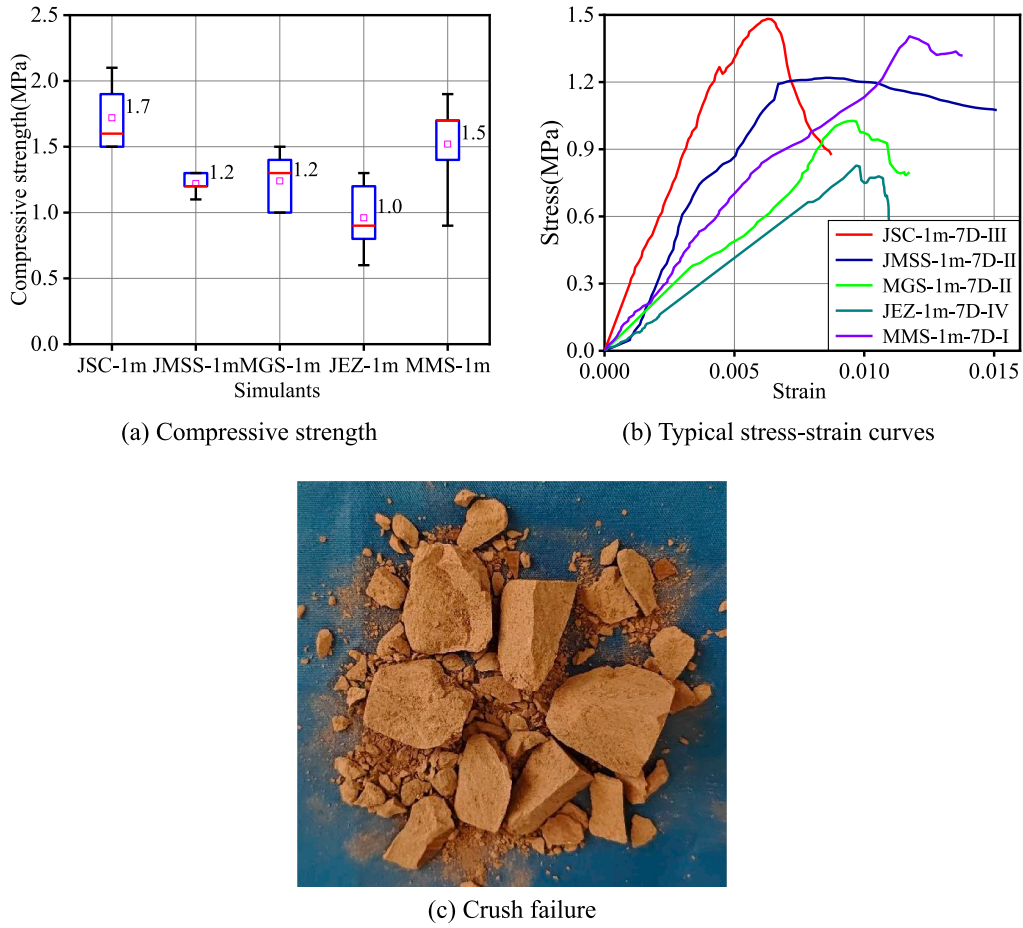


Fig. 5. Test results obtained from various Martian simulants.

3.2. Curing time

This section investigates the effect of ambient curing duration (at $\sim 20^\circ\text{C}$, 1 atm) on the baseline compressive strength development of the unprocessed Martian regolith simulant. Simulant JSC-1m, which exhibited the highest, albeit low, baseline strength among the tested compositions (Section 3.1), was selected for this evaluation.

The experimental results, presented in Fig. 6(a) and Table 2, indicate that extending the ambient curing time from 7 to 28 days has a minimal impact on compressive strength. The average strength increased only slightly from approximately 1.7 MPa at 7 days to around 1.8 MPa at 28 days. This marginal strength (f_u) gain with extended curing duration (D , in days) under these baseline conditions can be described by the following empirical relationship:

$$f_u = 0.01D + 1.63 \quad (1)$$

The equation mathematically represents the very gradual strength increase observed in the experimental data (Fig. 6(a)). Furthermore, examination of the stress-strain behavior (Fig. 6(b)) reveals no significant difference in material stiffness or overall mechanical response between specimens cured for 7, 14, or 28 days. This finding has a profoundly positive implication for Martian construction timelines. The stress-strain profiles remain comparable across the tested durations. Based on these findings, prolonging the ambient curing period beyond 7 days offers negligible improvement in the mechanical properties of the unprocessed JSC-1m simulant. Therefore, a 7-day ambient curing period was deemed sufficient for establishing consistent baseline properties and was adopted as a standard preparation step before applying further processing techniques (compaction, thermal treatment) investigated in subsequent sections.

3.3. Pressure

Compaction pressure is a crucial parameter in processing granular materials, as it directly influences particle packing density and interparticle contact, which are fundamental to developing mechanical strength [46–48]. Increased pressure promotes particle

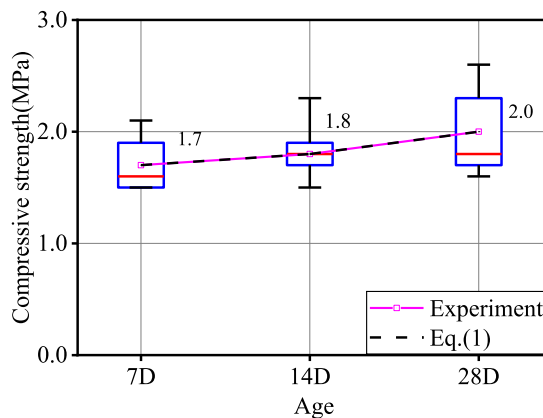
Table 2
Compressive test results.

Test ID	Specimens	Age (Days)	Compacting (MPa)	Water content (%)	Temperature (°C)	Particle size of SiO ₂ (μm)	f_u (MPa)	f_u -Mean (MPa)
001	JSC-1m-7D-I	7	0.1	35	20	1.00	1.9	1.7
002	JSC-1m-7D-II	7	0.1	35	20	1.00	2.1	
003	JSC-1m-7D-III	7	0.1	35	20	1.00	1.5	
004	JSC-1m-7D-IV	7	0.1	35	20	1.00	1.6	
005	JSC-1m-7D-V	7	0.1	35	20	1.00	1.5	1.2
006	JMSS-1m-7D-I	7	0.1	35	20	1.00	1.1	
007	JMSS-1m-7D-II	7	0.1	35	20	1.00	1.2	
008	JMSS-1m-7D-III	7	0.1	35	20	1.00	1.2	
009	JMSS-1m-7D-IV	7	0.1	35	20	1.00	1.3	1.2
010	JMSS-1m-7D-V	7	0.1	35	20	1.00	1.3	
011	MGS-1m-7D-I	7	0.1	35	20	1.00	1.4	
012	MGS-1m-7D-II	7	0.1	35	20	1.00	1.0	
013	MGS-1m-7D-III	7	0.1	35	20	1.00	1.0	1.2
014	MGS-1m-7D-IV	7	0.1	35	20	1.00	1.3	
015	MGS-1m-7D-V	7	0.1	35	20	1.00	1.5	
016	JEZ-1m-7D-I	7	0.1	35	20	1.00	1.2	
017	JEZ-1m-7D-II	7	0.1	35	20	1.00	0.9	1.0
018	JEZ-1m-7D-III	7	0.1	35	20	1.00	0.6	
019	JEZ-1m-7D-IV	7	0.1	35	20	1.00	0.8	
020	JEZ-1m-7D-V	7	0.1	35	20	1.00	1.3	
021	MMS-1m-7D-I	7	0.1	35	20	1.00	1.4	1.5
022	MMS-1m-7D-II	7	0.1	35	20	1.00	0.9	
023	MMS-1m-7D-III	7	0.1	35	20	1.00	1.9	
024	MMS-1m-7D-IV	7	0.1	35	20	1.00	1.7	
025	MMS-1m-7D-V	7	0.1	35	20	1.00	1.7	1.8
026	JSC-1m-14D-I	14	0.1	35	20	1.00	2.3	
027	JSC-1m-14D-II	14	0.1	35	20	1.00	1.5	
028	JSC-1m-14D-III	14	0.1	35	20	1.00	1.7	
029	JSC-1m-14D-IV	14	0.1	35	20	1.00	1.9	2.0
030	JSC-1m-14D-V	14	0.1	35	20	1.00	1.8	
031	JSC-1m-28D-I	28	0.1	35	20	1.00	2.3	
032	JSC-1m-28D-II	28	0.1	35	20	1.00	2.6	
033	JSC-1m-28D-III	28	0.1	35	20	1.00	1.7	3.4
034	JSC-1m-28D-IV	28	0.1	35	20	1.00	1.8	
035	JSC-1m-28D-V	28	0.1	35	20	1.00	1.6	
036	JSC-10MPa-35P-20C-1μ-I	7	10.0	35	20	1.00	2.9	
037	JSC-10MPa-35P-20C-1μ-II	7	10.0	35	20	1.00	3.5	4.0
038	JSC-10MPa-35P-20C-1μ-III	7	10.0	35	20	1.00	3.9	
039	JSC-10MPa-35P-20C-1μ-IV	7	10.0	35	20	1.00	3.1	
040	JSC-10MPa-35P-20C-1μ-V	7	10.0	35	20	1.00	3.4	
041	JSC-20MPa-35P-20C-1μ-I	7	20.0	35	20	1.00	4.0	4.9
042	JSC-20MPa-35P-20C-1μ-II	7	20.0	35	20	1.00	4.2	
043	JSC-20MPa-35P-20C-1μ-III	7	20.0	35	20	1.00	3.9	
044	JSC-20MPa-35P-20C-1μ-IV	7	20.0	35	20	1.00	3.8	
045	JSC-20MPa-35P-20C-1μ-V	7	20.0	35	20	1.00	4.0	4.6
046	JSC-40MPa-35P-20C-1μ-I	7	40.0	35	20	1.00	4.8	
047	JSC-40MPa-35P-20C-1μ-II	7	40.0	35	20	1.00	4.3	
048	JSC-40MPa-35P-20C-1μ-III	7	40.0	35	20	1.00	5.5	
049	JSC-40MPa-35P-20C-1μ-IV	7	40.0	35	20	1.00	5.3	4.4
050	JSC-40MPa-35P-20C-1μ-V	7	40.0	35	20	1.00	4.5	
051	JSC-60MPa-35P-20C-1μ-I	7	60.0	35	20	1.00	4.4	
052	JSC-60MPa-35P-20C-1μ-II	7	60.0	35	20	1.00	5.1	
053	JSC-60MPa-35P-20C-1μ-III	7	60.0	35	20	1.00	4.2	21.7
054	JSC-60MPa-35P-20C-1μ-IV	7	60.0	35	20	1.00	4.9	
055	JSC-60MPa-35P-20C-1μ-V	7	60.0	35	20	1.00	4.1	
056	JSC-40MPa-35P-20C-10μ-I	7	40.0	35	20	10.00	3.8	
057	JSC-40MPa-35P-20C-10μ-II	7	40.0	35	20	10.00	4.9	21.7
058	JSC-40MPa-35P-20C-10μ-III	7	40.0	35	20	10.00	4.6	
059	JSC-40MPa-35P-20C-10μ-IV	7	40.0	35	20	10.00	4.9	
060	JSC-40MPa-35P-20C-10μ-V	7	40.0	35	20	10.00	4.0	
061	JSC-40MPa-35P-20C-200n-I	7	40.0	35	20	0.20	20.9	21.7
062	JSC-40MPa-35P-20C-200n-II	7	40.0	35	20	0.20	26.2	
063	JSC-40MPa-35P-20C-200n-III	7	40.0	35	20	0.20	20.2	
064	JSC-40MPa-35P-20C-200n-IV	7	40.0	35	20	0.20	18.8	
065	JSC-40MPa-35P-20C-200n-V	7	40.0	35	20	0.20	22.5	21.7
066	JSC-40MPa-35P-20C-20n-I	7	40.0	35	20	0.02	19.2	
067	JSC-40MPa-35P-20C-20n-II	7	40.0	35	20	0.02	28.4	

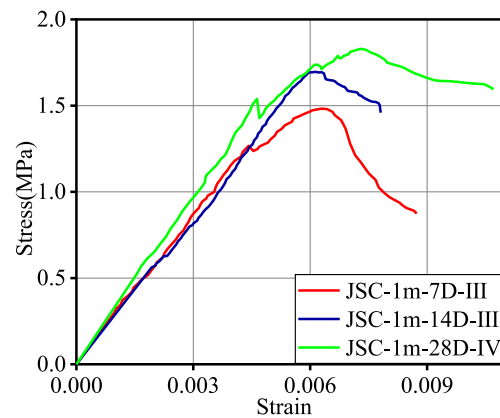
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Table 2 (continued)

Test ID	Specimens	Age (Days)	Compacting (MPa)	Water content (%)	Temperature (°C)	Particle size of SiO ₂ (μm)	f_u (MPa)	f_u -Mean (MPa)
068	JSC-40MPa-35P-20C-20n-III	7	40.0	35	20	0.02	24.0	22.9
069	JSC-40MPa-35P-20C-20n-IV	7	40.0	35	20	0.02	20.7	
070	JSC-40MPa-35P-20C-20n-V	7	40.0	35	20	0.02	22.4	
071	JSC-40MPa-35P-400C-200n-I	7	40.0	35	400	0.20	30.1	
072	JSC-40MPa-35P-400C-200n-II	7	40.0	35	400	0.20	25.8	
073	JSC-40MPa-35P-400C-200n-III	7	40.0	35	400	0.20	30.4	28.9
074	JSC-40MPa-35P-400C-200n-IV	7	40.0	35	400	0.20	29.9	
075	JSC-40MPa-35P-400C-200n-V	7	40.0	35	400	0.20	28.4	
076	JSC-40MPa-35P-600C-200n-I	7	40.0	35	600	0.20	40.7	
077	JSC-40MPa-35P-600C-200n-II	7	40.0	35	600	0.20	29.1	
078	JSC-40MPa-35P-600C-200n-III	7	40.0	35	600	0.20	27.4	30.9
079	JSC-40MPa-35P-600C-200n-IV	7	40.0	35	600	0.20	26.7	
080	JSC-40MPa-35P-600C-200n-V	7	40.0	35	600	0.20	30.7	
081	JSC-40MPa-35P-1000C-200n-I	7	40.0	35	1000	0.20	47.4	
082	JSC-40MPa-35P-1000C-200n-II	7	40.0	35	1000	0.20	48.0	
083	JSC-40MPa-35P-1000C-200n-III	7	40.0	35	1000	0.20	43.5	44.5
084	JSC-40MPa-35P-1000C-200n-IV	7	40.0	35	1000	0.20	41.1	
085	JSC-40MPa-35P-1000C-200n-V	7	40.0	35	1000	0.20	42.4	
086	JSC-40MPa-0P-20C-200n-I	7	40.0	0	20	0.20	0.4	
087	JSC-40MPa-0P-20C-200n-II	7	40.0	0	20	0.20	0.4	
088	JSC-40MPa-0P-20C-200n-III	7	40.0	0	20	0.20	0.4	0.4
089	JSC-40MPa-0P-20C-200n-IV	7	40.0	0	20	0.20	0.4	
090	JSC-40MPa-0P-20C-200n-V	7	40.0	0	20	0.20	0.4	
091	JSC-40MPa-15P-20C-200n-I	7	40.0	15	20	0.20	3.8	
092	JSC-40MPa-15P-20C-200n-II	7	40.0	15	20	0.20	5.0	
093	JSC-40MPa-15P-20C-200n-III	7	40.0	15	20	0.20	4.1	3.9
094	JSC-40MPa-15P-20C-200n-IV	7	40.0	15	20	0.20	2.1	
095	JSC-40MPa-15P-20C-200n-V	7	40.0	15	20	0.20	4.4	
096	JSC-40MPa-25P-20C-200n-I	7	40.0	25	20	0.20	19.8	
097	JSC-40MPa-25P-20C-200n-II	7	40.0	25	20	0.20	29.8	
098	JSC-40MPa-25P-20C-200n-III	7	40.0	25	20	0.20	27.9	25.3
099	JSC-40MPa-25P-20C-200n-IV	7	40.0	25	20	0.20	28.0	
100	JSC-40MPa-25P-20C-200n-V	7	40.0	25	20	0.20	20.8	
101	JSC-40MPa-40P-20C-200n-I	7	40.0	40	20	0.20	16.8	
102	JSC-40MPa-40P-20C-200n-II	7	40.0	40	20	0.20	20.0	
103	JSC-40MPa-40P-20C-200n-III	7	40.0	40	20	0.20	24.4	20.0
104	JSC-40MPa-40P-20C-200n-IV	7	40.0	40	20	0.20	17.6	
105	JSC-40MPa-40P-20C-200n-V	7	40.0	40	20	0.20	21.4	



(a) Compressive strength

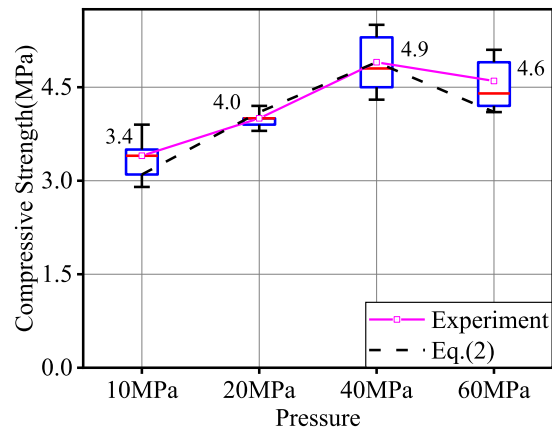


(b) Typical stress-strain curves

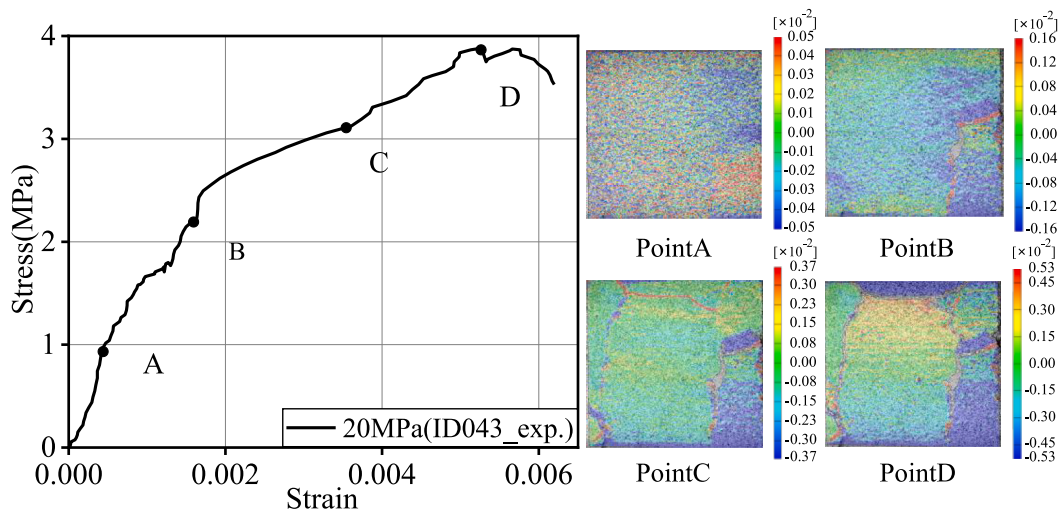
Fig. 6. Compressive strength and typical stress-strain curves for those having various curing ages.

rearrangement into a denser configuration, reduces bulk porosity, and potentially enhances interparticle bonding forces, possibly aided by capillary effects if minimal moisture is present [49]. This section investigates the effect of varying compaction pressure on the compressive strength and mechanical behavior of the ISRU bricks prepared from JSC-1m simulant.

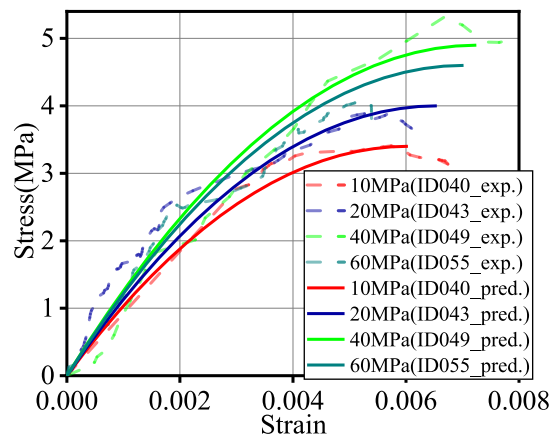
As illustrated in Fig. 7(a), the compressive strength of the bricks exhibited a marked improvement as compaction pressure increased



(a) Compressive strength



(b) A typical stress-strain curve



(c) Typical stress-strain curves

Fig. 7. Compressive strength and typical stress-strain curves obtained from various compacting pressures.

from 10 MPa to 40 MPa, rising from an average of 3.4 MPa–4.9 MPa. This significant enhancement correlates directly with the expected increase in material density and improved microstructure. However, further increasing the compaction pressure from 40 MPa to 60 MPa resulted in only marginal additional strength gains, suggesting diminishing returns beyond 40 MPa.

A representative stress-strain curve obtained from compressive testing is shown in Fig. 7(b). The behavior is characteristic of brittle materials, displaying an initial linear-elastic region (approximately OA), followed by progressive non-linearity associated with microcrack initiation and propagation (AB, BC), leading to peak stress and subsequent crushing failure (D). The overall shape of the stress-strain curve is qualitatively similar to that observed for conventional concrete. Consequently, the relationship between pressure and ultimate strength ($p-f_u$) as well as the corresponding stress-strain ($\sigma-\epsilon$) behavior can be articulated as follows:

$$f_u = -0.002p^2 + 0.16p + 1.70 \quad (2)$$

$$\sigma = f_u \left[2\epsilon / \epsilon_0 - (\epsilon / \epsilon_0)^2 \right] \quad (3)$$

$$E_c = 550\sqrt{f_u} \quad (4)$$

$$\epsilon_0 = 1.8f_u / E_c \quad (5)$$

As demonstrated in Fig. 7(c), these equations provide a good prediction of both the ultimate strength variation with pressure and the overall stress-strain behavior. Notably, higher compaction pressures lead to higher strengths (f_u) and, consequently, stiffer materials (higher E_c), reflected in the steeper initial slopes of the experimental stress-strain curves. Microstructural examination supports these findings. Fig. 8(a) (lower pressure) reveals a relatively porous microstructure with less intimate particle contact, hindering effective load transfer. In contrast, Fig. 8(b) (higher pressure) shows a significantly denser matrix with reduced porosity and closer particle packing. This densification facilitates more efficient stress distribution and stronger interparticle interactions, directly contributing to the higher observed compressive strength (4.9 MPa vs. 3.4 MPa). Considering the substantial strength increase up to 40 MPa and the negligible improvement observed upon further increasing pressure to 60 MPa, a compaction pressure of 40 MPa is identified as an optimal level for this material system under the tested conditions. It provides a favorable balance between achieved strength enhancement and the practicalities of applying high pressure. Therefore, 40 MPa was selected as the standard compaction pressure for subsequent investigations.

3.4. Aggregate size

The sourcing of such fine-grained material is a critical consideration for ISRU. Fortunately, the Martian surface is rich in fine dust, with particle size distributions observed to follow a power law that includes a significant nano-scale fraction [50]. The particle size distribution of the primary aggregate significantly influences the packing density and resultant strength of compacted materials. This section compares the effect of utilizing nano-scale SiO₂ (specifically, 200 nm particles) versus micron-sized SiO₂ (specifically, ~1 μ m particles) as the main constituent in ISRU bricks prepared via compaction (40 MPa) at ambient temperature (~20 °C), using the JSC-1m simulant base composition.

The experimental results, presented in Fig. 9(a), reveal a dramatic difference in compressive strength based on the SiO₂ particle size. Bricks incorporating nano-SiO₂ (200 nm) achieved average compressive strengths exceeding 20 MPa under these conditions. In stark contrast, specimens fabricated with micron-sized SiO₂ exhibited substantially lower strengths, typically below 5 MPa (consistent with results in Section 3.3). This significant strength enhancement associated with nano-SiO₂ can be primarily attributed to physical effects related to particle packing and surface area: 1) nano-sized particles can pack much more densely than micron-sized ones, significantly reducing the volume of interstitial voids within the compacted matrix (compare Fig. 10(a) with Fig. 10(b)). This denser

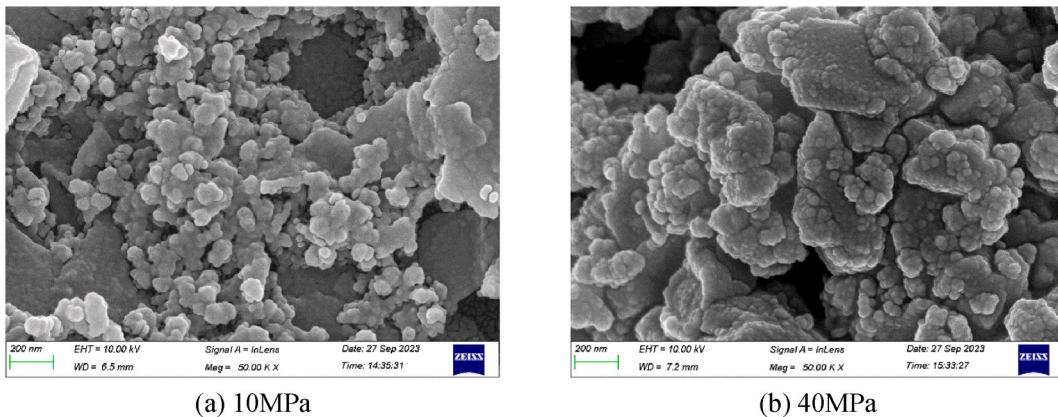


Fig. 8. SEM comparisons of specimens made by various compacting pressures.

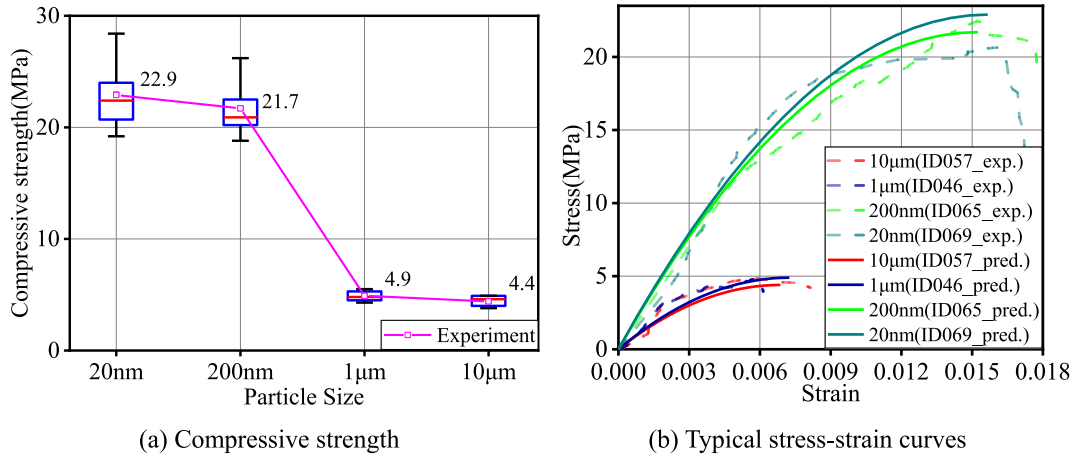


Fig. 9. Compressive strength and typical stress-strain curves obtained from various particle sizes.

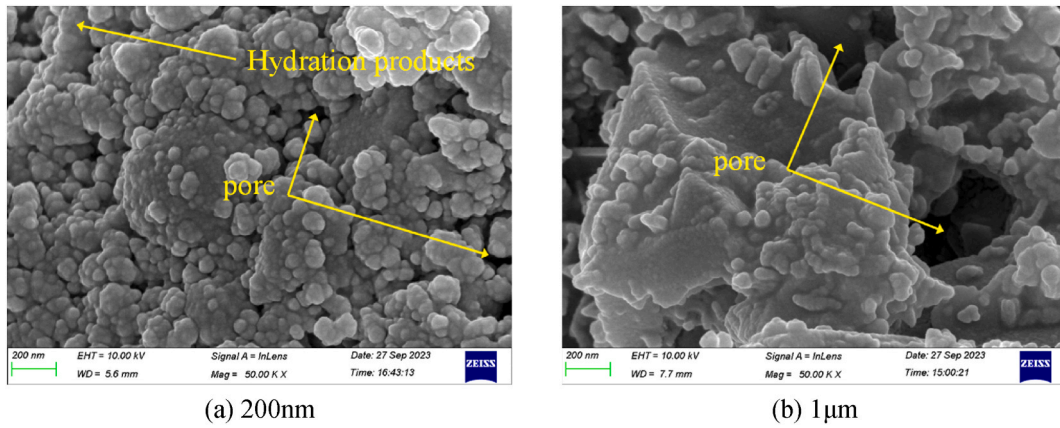


Fig. 10. SEM comparisons of specimens made by various particle sizes.

structure provides more effective stress transfer paths and reduces defect sites where failure can initiate; 2) The vastly larger total surface area of nano-particles enhances interparticle forces (e.g., capillary bridging if moisture is present) and provides significantly more contact points for potential bonding.

Microstructural analysis (Fig. 10) visually confirms these points. Fig. 10(a) (nano-SiO₂) shows a markedly denser, more homogeneous microstructure with finer porosity compared to Fig. 10(b) (micron-SiO₂), which displays larger voids and a less tightly packed structure. While some reaction or enhanced bonding at particle contacts might occur even at ambient temperature under pressure [51, 52], the primary benefit observed here stems from the superior packing and increased contact facilitated by the nano-scale particles.

The substantial strength increase achieved by using nano-SiO₂ can be empirically represented by introducing a scaling factor into the previously established pressure-strength relationship (Eq. (2)):

$$f_u = \begin{cases} 4.79 * Eq(2) & \text{if nano SiO}_2 \text{ is applied} \\ Eq(2) & \text{if micron - sized SiO}_2 \text{ is applied} \end{cases} \quad (6)$$

As shown in Fig. 9(b), this modified relationship accurately captures the strength difference, and consequently, predicts the significantly stiffer stress-strain response observed for the nano-SiO₂ specimens, consistent with the constitutive model (Eqs. (3)–(5)).

Given the profound positive impact on compressive strength, achieving values over 20 MPa even without high-temperature treatment, the use of 200 nm SiO₂ particles is identified as highly advantageous for producing high-performance ISRU bricks. This particle size was therefore selected for further investigation involving thermal treatment. It is important to note that this investigation of the particle size effect was conducted specifically with the JSC-1m simulant. While the dramatic strength enhancement observed is believed to be primarily driven by the physical mechanisms of improved particle packing and increased surface area, it is conceivable that the magnitude of this effect could vary with different simulant compositions. The fundamental principles of densification should apply to other silicate-based Martian simulants, and it was hypothesized that a similar positive trend would be observed.

3.5. Temperature

Subsequent thermal treatment (sintering) following compaction can significantly enhance the mechanical properties of particulate materials by promoting interparticle bonding, densification, and porosity reduction through mechanisms like solid-state diffusion or viscous flow, depending on the material composition and temperature. This section evaluates the effect of post-compaction thermal treatment temperature on the ISRU bricks previously prepared using nano-SiO₂ (200 nm) and compacted at 40 MPa. Specimens were heated to target temperatures ranging from 400 °C to 1000 °C and held for 6 h before controlled cooling.

The results, presented in Fig. 11(a) and Table 2, demonstrate a substantial increase in average compressive strength with increasing treatment temperature. Starting from the baseline strength of 21.7 MPa for specimens processed only at ambient temperature (20 °C), the strength progressively increased to 28.9 MPa (400 °C), 30.9 MPa (700 °C), and ultimately reached 44.5 MPa after treatment at 1000 °C. This represents more than a doubling of the strength compared to the unheated, compacted material. Within the tested range (20 °C–1000 °C), the relationship between the ultimate compressive strength (f_u) and the treatment temperature (T , in °C) can be described by the following empirical linear equation, which builds upon the strength achieved after compaction (Eq. (6)):

$$f_u = Eq.(6) + 0.02T - 2.00 \quad (7)$$

Microstructural analysis provides clear evidence for the mechanisms driving this strength enhancement. Comparing the microstructure of a specimen treated at 1000 °C (Fig. 12) with one compacted at ambient temperature (Fig. 10(a)), reveals significant densification and porosity reduction at the higher temperature. Evidence of particle fusion and necking between adjacent SiO₂ particles, characteristic of sintering, is apparent in Fig. 12, creating a more monolithic and structurally sound matrix capable of sustaining higher loads.

Consistent with the increased strength, the thermal treatment also results in significantly stiffer materials, as indicated by the steeper initial slopes of the stress-strain curves shown in Fig. 11(b). This aligns with the constitutive model (Eqs. (3)–(5)), where a higher ultimate strength (f_u) naturally leads to a higher calculated elastic modulus (E_c). The predictive model, combining Eq. (7) with Eqs. (3)–(5), shows good agreement with the experimental stress-strain responses across the investigated temperature range (Fig. 11(b)).

These findings clearly demonstrate that post-compaction thermal treatment, particularly up to 1000 °C, is a highly effective method for dramatically enhancing the compressive strength and stiffness of the nano-SiO₂ based ISRU bricks, transforming the compacted powder into a high-performance structural material.

3.6. Water content

The role of initial water content (WC) is critical to the bonding mechanism, serving a dual function as both a processing lubricant and a chemical reactant. This section investigates its influence on the final properties to explicitly differentiate its function from that of a conventional hydration process. ISRU bricks were prepared using nano-SiO₂ (200 nm), compacted at 40 MPa, with initial WC varying from 0 % to 40 % by weight, and tested after ambient temperature (~20 °C) curing.

The experimental results, shown in Fig. 13(a), demonstrate a profound dependence of compressive strength on the initial WC. Strength increases sharply from a negligible 0.4 MPa for the dry mix (0 % WC) to approximately 25.3 MPa at 25 % WC. Beyond this point, increasing the initial WC up to 40 % results in only marginal strength improvements. This trend indicates a complex relationship where water first acts as a lubricant, enabling particles to rearrange into a dense matrix under pressure. Microstructural evidence confirms this lubricating effect: the highly porous structure of the zero-WC specimen (Fig. 14), indicative of poor compaction, contrasts sharply with the much denser matrix achieved when sufficient water is present (Fig. 10(a)).

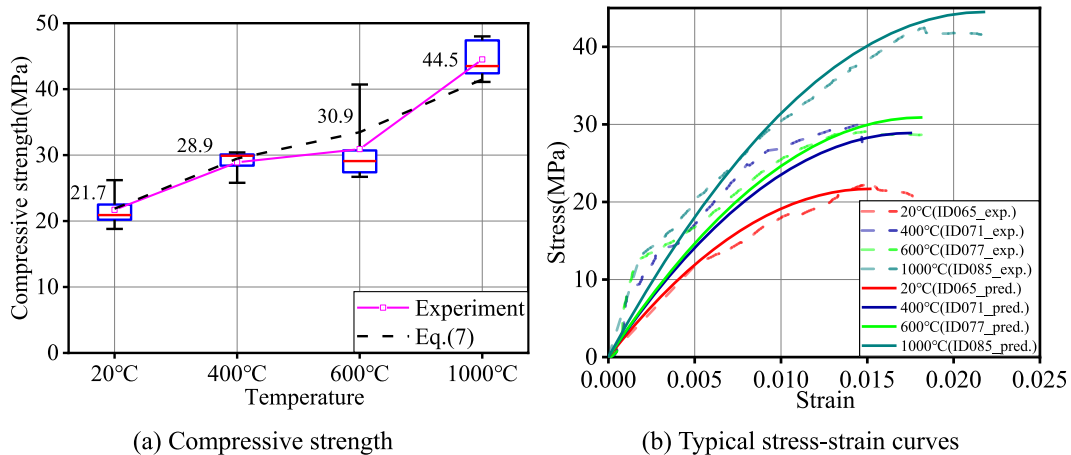


Fig. 11. Compressive strength and typical stress-strain curves obtained from various temperatures.

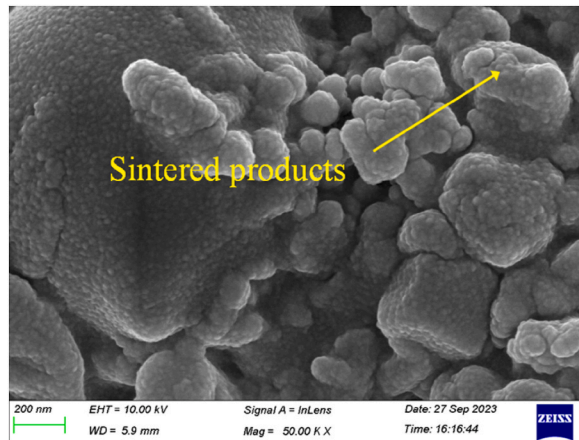


Fig. 12. SEM observations of specimens cured at 1000 °C.

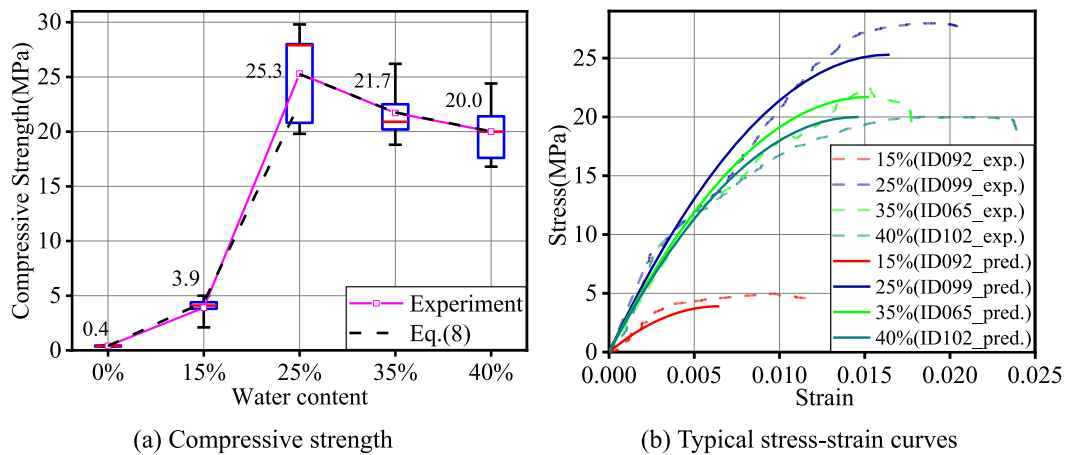


Fig. 13. Compressive strength and typical stress-strain curves obtained from various water contents.

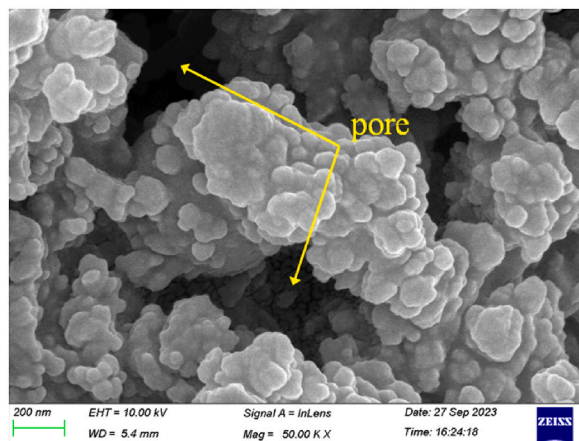


Fig. 14. SEM observation of specimen cured at water-free condition.

However, beyond simple physical densification, the applied pressure is hypothesized to create localized high-pressure points at particle contacts, facilitating a phase change or creating conditions where water can react with the high-surface-area nano-SiO₂ particles. This pressure-facilitated hydration creates chemical bonds that supplement the physical particle interlocking, contributing

significantly to the material's final strength. The negligible strength gain beyond 25 % WC suggests an optimal balance has been reached between the water needed for lubrication and the amount that can be effectively utilized in these localized reactions under 40 MPa of pressure. Microstructural analysis (Fig. 10(a)) confirms the formation of a dense matrix, with features identified as hydration products visibly cementing the nano-particles together, providing direct evidence for this chemical bonding component.

An initial water content of 35 % is therefore recommended, as it ensures the plateau of high strength is reliably reached while also

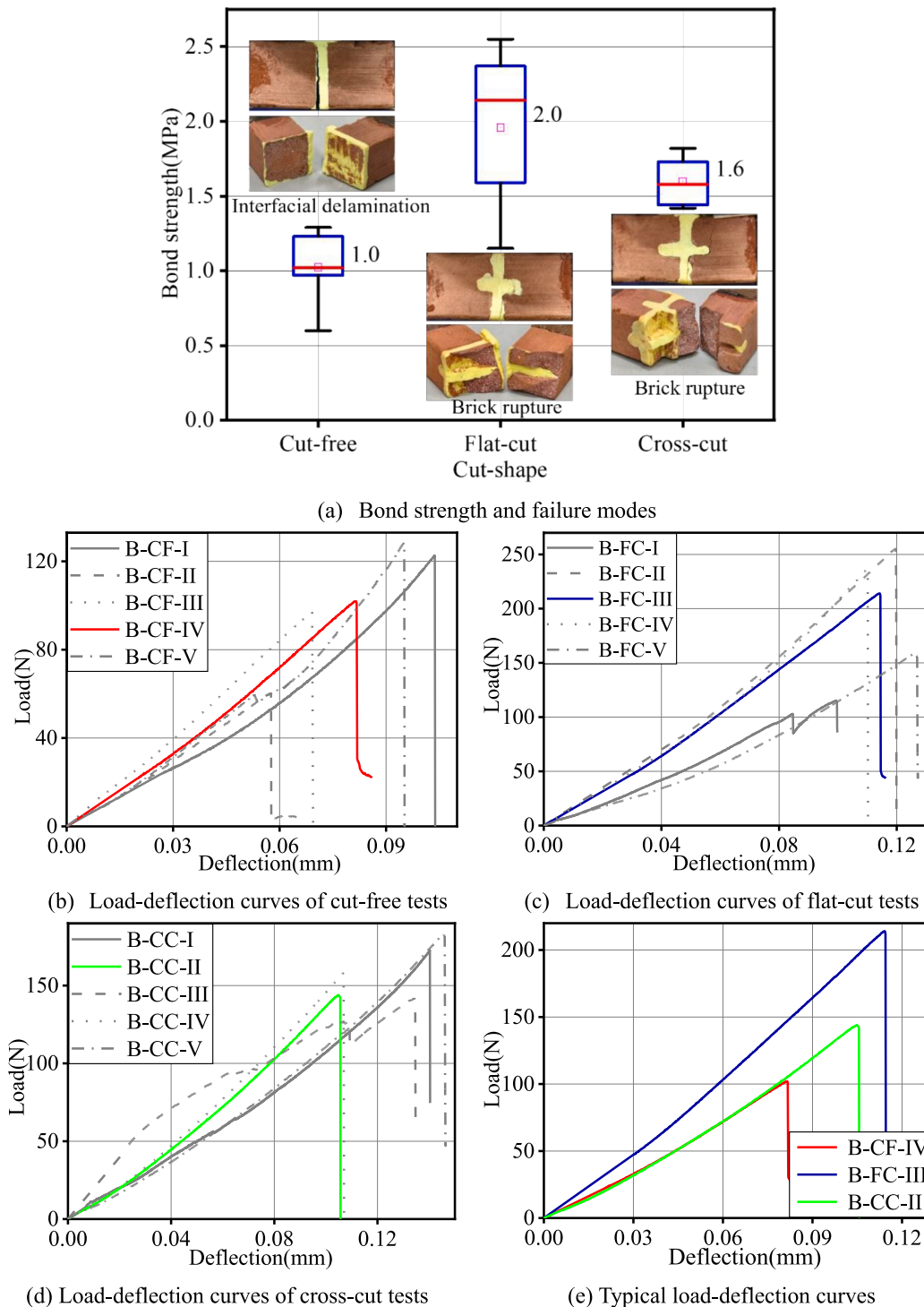


Fig. 15. Experimental results of bond tests.

improving the mixture's handling characteristics (workability) prior to compaction. The complex relationship between initial water content (w , as a fraction) and compressive strength (f_u) for the nano-SiO₂, 40 MPa, 20 °C condition can be empirically modeled using the following piecewise function:

$$f_u = \begin{cases} \text{Eq. (7)} + 0.38e^{16.40w} - 21.87 & w < 25\% \\ \text{Eq. (7)} - 34.89w + 12.10 & w \geq 25\% \end{cases} \quad (8)$$

Equation (8) models the strength variation due to this synergistic physical-chemical effect, building upon the strength predicted by the temperature model (Eq. (7)). As shown in Fig. 13(a), this model captures the observed trend well. Furthermore, using the strengths predicted by Eq. (8) within the constitutive model (Eqs. (3)–(5)) allows for accurate prediction of the corresponding stress-strain behaviors, as demonstrated by the good agreement with experimental curves in Fig. 13(b). These results underscore the unique bonding mechanism, which leverages pressure to facilitate localized hydration, thereby achieving high strength with minimal water, a critical advantage for Martian construction.

3.7. Sulfur-based bond

Effective assembly of the prefabricated ISRU bricks requires a suitable bonding agent. Building upon previous studies utilizing molten elemental sulfur [3], this section evaluates the performance of sulfur mortar for bonding the developed ISRU bricks and investigates the critical influence of interface preparation on improving bond strength.

ISRU bricks fabricated under optimized compaction conditions (nano-SiO₂ [200 nm], 40 MPa pressure, ambient temperature ~20 °C, resulting nominal compressive strength ~22 MPa) were used. Pairs of bricks were bonded together using molten elemental sulfur. Three distinct interface conditions were tested. The 'Cut-free' or unprepared specimens represented the baseline condition, using the as-molded surfaces of the bricks without any modification. For 'Flat-cut' specimens, approximately 3 mm of material was removed from the bonding face using a diamond-bladed masonry saw to create a fresh, uniform plane. For 'Cross-cut' specimens, after the initial flat cut, a grid of perpendicular grooves was machined into the surface to create a textured, keyed interface as depicted in Fig. 4(b).

The interface preparation method had a profound effect on both the bond strength and the failure mode, as summarized in Fig. 15 (a) and Table 3. Unprepared interfaces consistently failed via interfacial delamination between the sulfur mortar and the brick surface, achieving a relatively low average bond strength of approximately 1.0 MPa. This suggests poor adhesion or mechanical keying on the original molded surface. In contrast, both flat-cut and cross-cut interfaces predominantly exhibited failure via rupture of the ISRU brick material itself, rather than failure at the bond line. This critical shift in failure mode indicates that the bond strength achieved with these preparations exceeded the flexural strength of the brick material under the test conditions. The apparent average bond strengths corresponding to these brick failures were significantly higher: 2.0 MPa for the flat-cut interface and 1.6 MPa for the cross-cut interface. The enhanced performance with prepared surfaces is likely attributable to improved mechanical interlocking and an increased effective bonding area created by the cutting process, leading to a stronger joint.

Analysis of the load-deflection curves (Fig. 15(b)–(d)) revealed generally linear behavior up to the peak load for all interface types, indicating predictable loading responses before failure. Representative curves comparing the conditions (Fig. 15(e)) clearly illustrate the substantial improvements in both load-carrying capacity and stiffness achieved with the flat-cut and cross-cut preparations compared to the unprepared interface. Comparing the two effective preparation methods, the flat-cut interface provided the highest average bond strength (2.0 MPa). Furthermore, as detailed in Table 3, this preparation method consistently yielded bond performance where the failure initiated in the brick, with a minimum implied tensile capacity reliably exceeding 1.2 MPa. Considering its relative ease of implementation compared to cross-cutting, the flat-cut preparation method is identified as the most advantageous approach,

Table 3
Bond test results.

Test ID	Specimens	Cut-shape	Failure modes	f_u (MPa)	f_u -Mean (MPa)
01	B-CF-I	Cut-free	Interfacial delamination	1.2	1.0
02	B-CF-II	Cut-free	Interfacial delamination	0.6	
03	B-CF-III	Cut-free	Interfacial delamination	1.0	
04	B-CF-IV	Cut-free	Interfacial delamination	1.0	
05	B-CF-V	Cut-free	Interfacial delamination	1.3	
06	B-FC-I	Flat-cut	Brick rupture	1.2	2.0
07	B-FC-II	Flat-cut	Brick rupture	2.6	
08	B-FC-III	Flat-cut	Brick rupture	2.1	
09	B-FC-IV	Flat-cut	Brick rupture	2.4	
10	B-FC-V	Flat-cut	Brick rupture	1.6	
11	B-CC-I	Cross-cut	Brick rupture	1.7	1.6
12	B-CC-II	Cross-cut	Brick rupture	1.4	
13	B-CC-III	Cross-cut	Brick rupture	1.4	
14	B-CC-IV	Cross-cut	Brick rupture	1.6	
15	B-CC-V	Cross-cut	Brick rupture	1.8	

*"B" Bond test, "CF" Cut-free, "FC" Flat-cut, "CC" Cross-cut.

offering a practical means to achieve robust and reliable sulfur mortar joints for assembling ISRU brick structures on Mars.

While these results demonstrate high initial bond strength, long-term durability in the Martian environment is critical. The primary environmental challenges are extreme thermal cycling and high UV radiation. The large diurnal temperature swings could induce fatigue cracking due to a mismatch in thermal expansion coefficients between the sulfur mortar and the silicate bricks, as well as

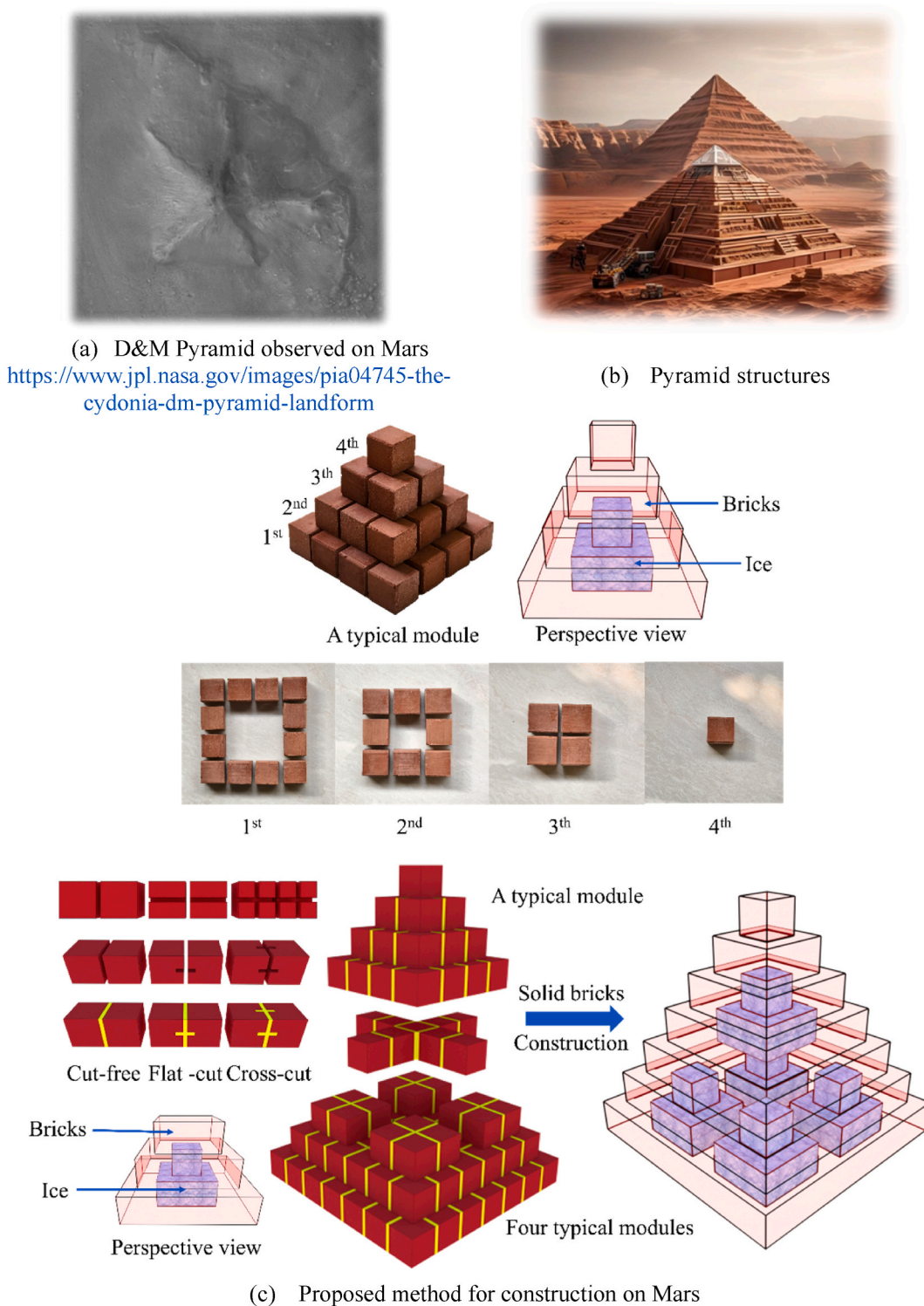


Fig. 16. Construction of pyramid-shaped structures on Mars.

internal stresses from sulfur's own solid-state phase transition. However, using sulfur as a thin mortar layer, as done here, inherently minimizes the magnitude of these stresses compared to bulk sulfur concrete. The structural bricks themselves will also provide significant thermal mass, dampening the temperature extremes experienced by the joints. For UV radiation, the risk is negligible; as the mortar is used exclusively within the joints, the opaque ISRU bricks provide complete shielding from solar radiation, preventing potential photodegradation. Future work could further enhance thermomechanical performance by investigating the use of sulfur modifiers (plasticizers) to suppress the phase transition and improve ductility at low temperatures.

4. Construction and design of Martian masonry

The concept integrates prefabricated high-strength ISRU bricks, sulfur mortar bonding, and a modular pyramid geometry (inspired by structures like Fig. 16 (a)) to create robust habitats (Fig. 16 (b)) suited for the Martian environment. While other geometries, such as a dome, offer superior internal volume-to-surface-area ratios and are theoretically optimal for managing pressurization via pure membrane tension, they pose significant construction challenges for a masonry-based system. A pyramid, in contrast, is exceptionally well-suited to assembly from discrete blocks. Its geometry is inherently stable under compression and allows for simpler, corbelled construction techniques that minimize the need for the complex formwork a masonry dome would require. Furthermore, for a pressurized habitat made of massive masonry, the pyramid form is highly effective. The substantial self-weight of the sloped walls acts to counteract the tensile forces induced by internal pressure, keeping the structural elements primarily in a state of compression, which is ideal for the developed ISRU brick material. Therefore, the pyramid is not chosen for volumetric efficiency, but as a robust and practical geometry that leverages the specific strengths of a compression-dominant, modular masonry construction paradigm in the Martian context.

4.1. Construction process

The proposed methodology emphasizes modularity and leverages the findings from the material development sections:

- 1) Prefabrications: High-strength ISRU bricks are manufactured using optimized compaction (e.g., 40 MPa) and potentially thermal treatment, depending on desired strength and energy availability (Sections 3.3-3.5).
- 2) On-Site assembly: Prefabricated bricks are assembled on Mars using molten sulfur mortar. Critically, brick interfaces should be prepared (e.g., flat-cut) to ensure robust bond strength (>1.2 MPa minimum capacity) as determined in Section 3.7.
- 3) In-situ supports: For creating internal cavities or complex geometries within modules (Fig. 16 (c)), temporary internal supports, potentially utilizing readily available Martian resources like water ice, could be employed. The ice would later be sublimated to create the internal space.
- 4) Scalability: Standardized pyramid modules can be constructed and potentially interconnected (Fig. 16 (c)) to create larger, expandable habitat complexes.

A key aspect of this construction strategy is the strategic placement of bricks based on their manufacturing process and resultant durability. While bricks produced via ambient-temperature compaction possess sufficient strength for primary structural roles, the exterior surfaces of the habitat demand superior erosion resistance. Therefore, it is proposed that the outermost cladding of the pyramid structure be constructed using bricks that have undergone high-temperature sintering (1000°C). As established in Section 3.5, these sintered bricks exhibit ceramic-like properties, providing the necessary hardness to withstand long-term abrasion from Martian dust storms, thereby ensuring the habitat's longevity.

Furthermore, transitioning from laboratory fabrication to large-scale manufacturing on Mars necessitates a scalable and automated approach. The proposed production method is envisioned as a semi-continuous, automated process powered by solar arrays. This system would involve: 1) robotic excavation of regolith, 2) automated mixing of the processed regolith with a precise amount of water, 3) conveyance of the mixture into molds, and 4) transfer of the molds into a dedicated hydraulic or mechanical press for compaction. After pressing, robotic arms would eject the finished bricks and transport them to a curing area or, for exterior-grade bricks, into a sintering furnace. An essential consideration in this process is the energy budget. The primary energy sinks are the mechanical work for compaction and the thermal energy for optional sintering. A significant advantage of this method is its low-energy baseline pathway; producing structurally competent bricks (~ 22 MPa) via compaction at ambient temperature is considerably less power-intensive than alternatives relying on bulk melting of regolith or the high temperatures required for traditional cement production. The energy-intensive sintering process would be employed strategically, reserved only for high-performance components like exterior cladding, thereby ensuring the overall power budget remains practical for the energy-constrained Martian environment.

This approach, decoupling brick manufacturing from final assembly, potentially offers enhanced quality control, construction speed, and maintainability.

4.2. FE design

To assess the structural viability of the concept, FEA was performed to simulate the behavior of a representative pyramid module under Martian conditions. The primary objective was to determine the stress distribution within the structure under construction (self-weight) and operational (self-weight + internal pressure) loads, and to verify if these stresses remain within the capacity of the ISRU brick material and the sulfur mortar joints.

A four-level pyramid module was modeled (initial levels shown in Fig. 17 (a)). Each brick was designated to have a width b and length l of 1 m, and each floor has a height of 3 m, with the thickness of the top layer measuring 0.3 m. In the field, the substantial mass of Martian bricks, exemplified by the 1 m³ cube elements, can be attained through the sulfur bonding of multiple smaller bricks. Geometric and material symmetry were utilized, modeling one half of the structure. ANSYS Solid65 elements, suitable for brittle materials like concrete and capable of simulating cracking in tension and crushing in compression, were employed to represent the ISRU brick masonry. Material properties implemented in the FE model were directly based on the experimental findings. The average density was 2.51 g/cm³, as detailed in Table 4. The non-linear stress-strain relationship was defined using the constitutive model described by Eqs. (3)–(5). An ultimate compressive strength (f_u) of 22 MPa was used, representing the experimentally determined strength for nano-SiO₂ bricks compacted at 40 MPa without high-temperature thermal treatment (Section 3.4, Fig. 9 (a)). The tensile cracking strength of the material (representing potential failure either in the brick or the bond) was set to 1.2 MPa. This value corresponds to the minimum reliable bond strength capacity observed in the four-point bending tests for the recommended flat-cut sulfur mortar interface (Section 3.7, Table 3), which consistently forced failure into the brick material itself. This assumes the bond interface is the critical link under tension.

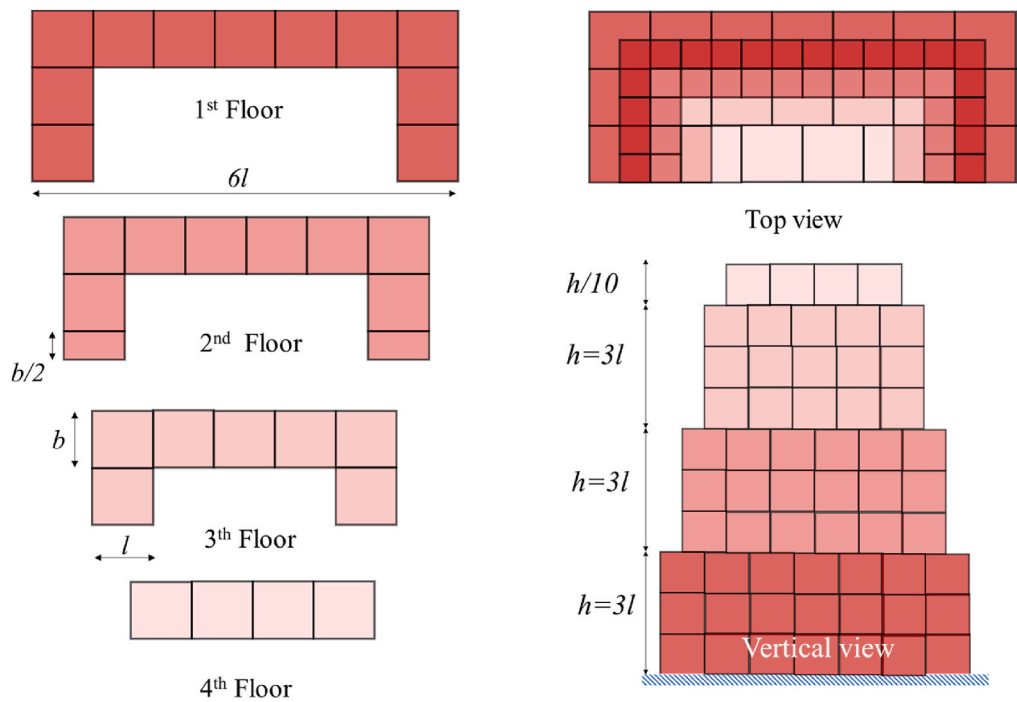
It must be emphasized that this deterministic FEA serves as a proof-of-concept and relies on mean or conservative-minimum values from the experimental program. A comprehensive structural design would necessitate a full sensitivity and uncertainty analysis to quantify the effects of material property variations—such as deviations in brick compressive strength, bond strength, and density—on the habitat's stability. Such a probabilistic assessment was beyond the scope of this initial feasibility study but is a critical next step. Following studies should incorporate parametric analyses to establish performance boundaries and inform the development of reliability-based design factors appropriate for mission-critical infrastructure. It is crucial to acknowledge that these strength values were obtained from small-scale laboratory specimens. Due to the statistical size effect inherent in quasi-brittle materials, where larger volumes have a higher probability of containing strength-limiting flaws, the nominal strength of full-scale structural components is expected to be lower. Therefore, the present FEA should be interpreted as a proof-of-concept validation of the structural form. A formal design protocol for a mission-critical application would require these characteristic strength values to be scaled down by a size-dependent reduction factor to ensure a conservative and safe design. Furthermore, it is crucial to acknowledge that the material properties used in this FEA are defined by empirical models (Eqs. (3)–(5)) developed from tests on small-scale laboratory specimens. A conservative design philosophy must therefore be adopted. Firstly, due to the statistical size effect inherent in quasi-brittle materials, where larger volumes have a higher probability of containing strength-limiting flaws, the nominal strength of full-scale structural components is expected to be lower than the values reported here. A formal design protocol would require the characteristic strength to be scaled down by a size-dependent reduction factor. Secondly, the phenomenological models themselves, while well-fitted to the current data, are based on a limited experimental set and their generalizability is not yet established. The present simulation should therefore be viewed as a validation of the structural concept rather than a definitive design analysis.

The base of the pyramid model was fully fixed. Symmetry boundary conditions were applied on the plane of symmetry (Fig. 17 (b)). Two primary load cases were analyzed using a non-linear solver. In the construction stage, self-weight under Martian gravity (3.71 m/s²) was applied. Then, self-weight (Martian gravity) combined with an internal pressure differential of 101.3 kPa (representing a 1 atm internal habitat pressure against the near-vacuum exterior), applied as an outward pressure on the internal surfaces of the module for the operational stage (Fig. 17 (c)).

Stress contours for maximum principal tensile and compressive stress are shown in Figs. 18 and 19. In the construction stage, stresses were low, with maximum principal tensile stress $\approx$ 0.04 MPa and maximum principal compressive stress $\approx$ 0.13 MPa (Figs. 18 and 19). The internal pressure significantly increased stresses during the operational stage. The maximum principal tensile stress reached 1.15 MPa, localized near an internal corner on the second level (Fig. 18 (c)). The maximum principal compressive stress was approximately 0.59 MPa (Fig. 19 (c)). The critical result is the maximum principal tensile stress of 1.15 MPa predicted under operational conditions. This value is below the 1.2 MPa tensile cracking strength adopted in the model, which represents the minimum reliable capacity of the flat-cut sulfur mortar bond joint with the proposed bricks derived experimentally. Therefore, the FEA results indicate that the proposed pyramid structure, constructed using the developed ISRU bricks (even those produced without high heat) and the optimized sulfur bonding technique, possesses sufficient structural integrity to withstand the combined effects of Martian gravity and a 1 atm internal pressure differential.

It should be noted that this analysis focuses on the primary static loads, which are foundational for establishing structural viability. A comprehensive design would also need to account for other potential loads, including aeolian forces from dust storms, operational live loads, seismic events, and thermal stresses from diurnal temperature swings. However, a detailed quantitative simulation of wind and seismic loads is considered premature. High-fidelity engineering models require site-specific atmospheric and ground motion data, which remains sparse for Mars despite ongoing data collection from missions like InSight. Qualitatively, the pyramid geometry is inherently advantageous for these conditions; its low-profile, sloped surfaces reduce wind pressure, and its wide base provides stability against ground shaking. Further analysis of these dynamic and thermal loads constitutes a critical next step in design maturation, contingent on the availability of more detailed environmental data.

Finally, it is acknowledged that the experimental fabrication was conducted under terrestrial atmospheric pressure and gravity. While the low Martian atmospheric pressure affects the stability of liquid water, the envisioned automated manufacturing process would occur in a contained system, mitigating rapid evaporation during the brief mixing and compaction cycle. Similarly, the influence of reduced gravity on the final compacted material properties is considered minor relative to the high mechanical pressure (40 MPa), which is the dominant consolidation force. The current study establishes a crucial baseline; however, future work should include validating material performance when fabricated within high-fidelity environmental chambers that simulate Martian conditions.



(a) Layout of the 1/2 numerical model

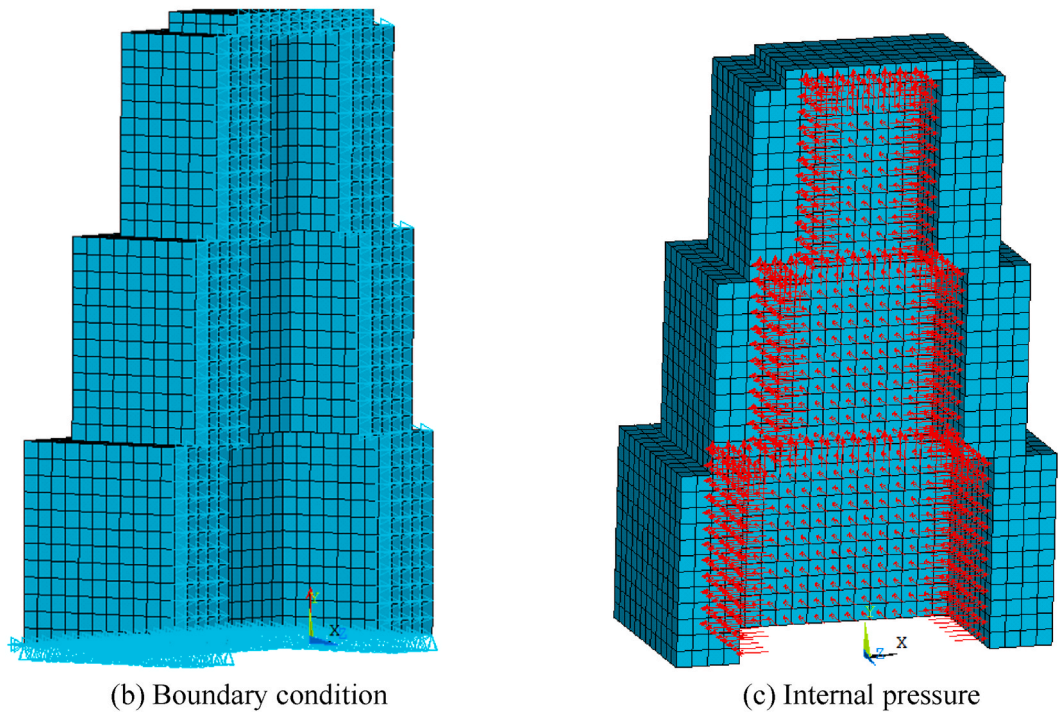


Fig. 17. Modeling details.

Table 4
Density measurement results.

Experiment Number	Mass (g)	Volume (cm ³)	Density (g/cm ³)	Density-Mean (g/cm ³)
1	411.28	162.56	2.53	2.51
2	403.92	162.87	2.48	
3	404.93	163.94	2.47	
4	418.56	163.50	2.56	
5	409.96	163.33	2.51	

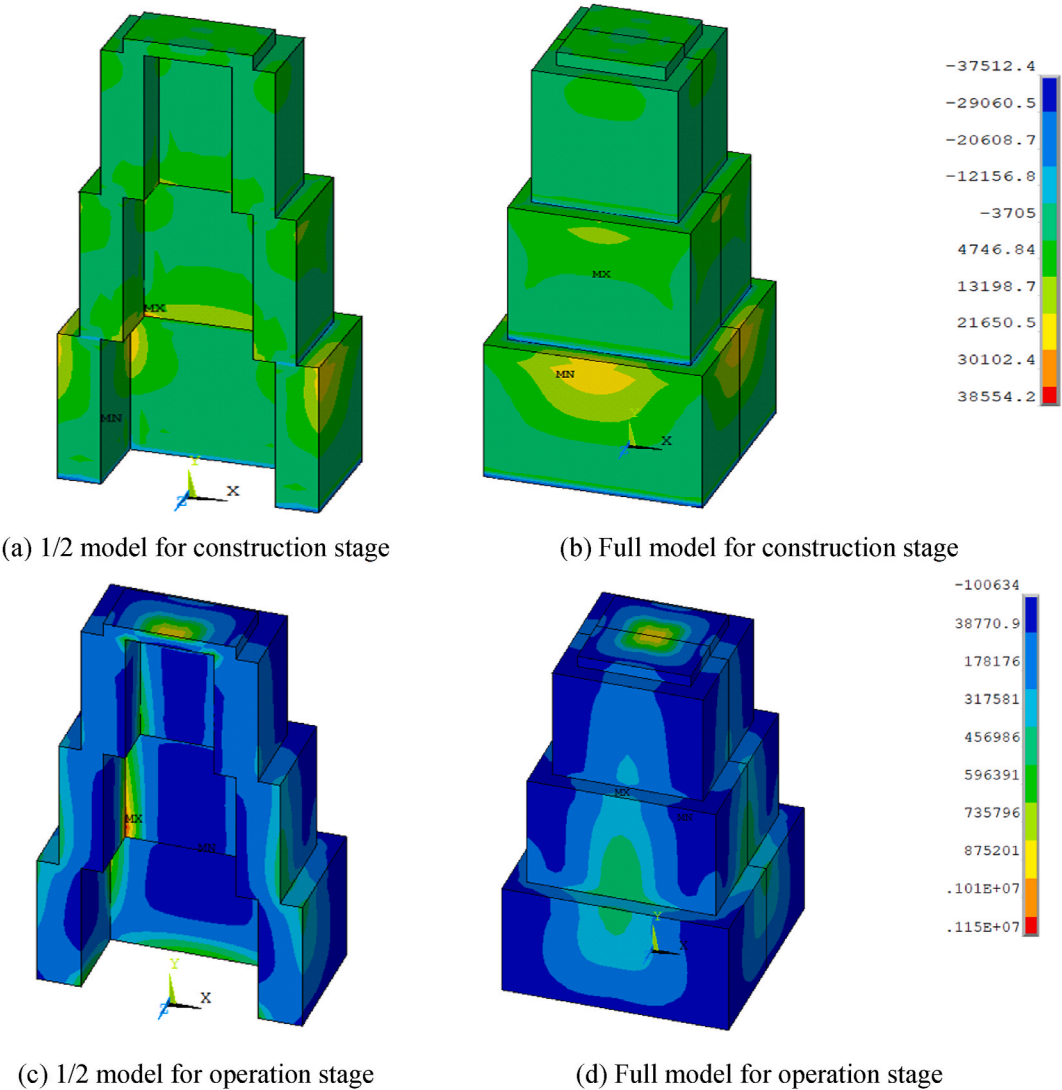


Fig. 18. Principal tensile stress of two scenarios.

5. Conclusions

This research successfully developed and validated an integrated approach for Martian habitat construction, encompassing ISRU material development, optimized bonding techniques, modular assembly, and FE verification. The key findings and contributions are summarized as follows:

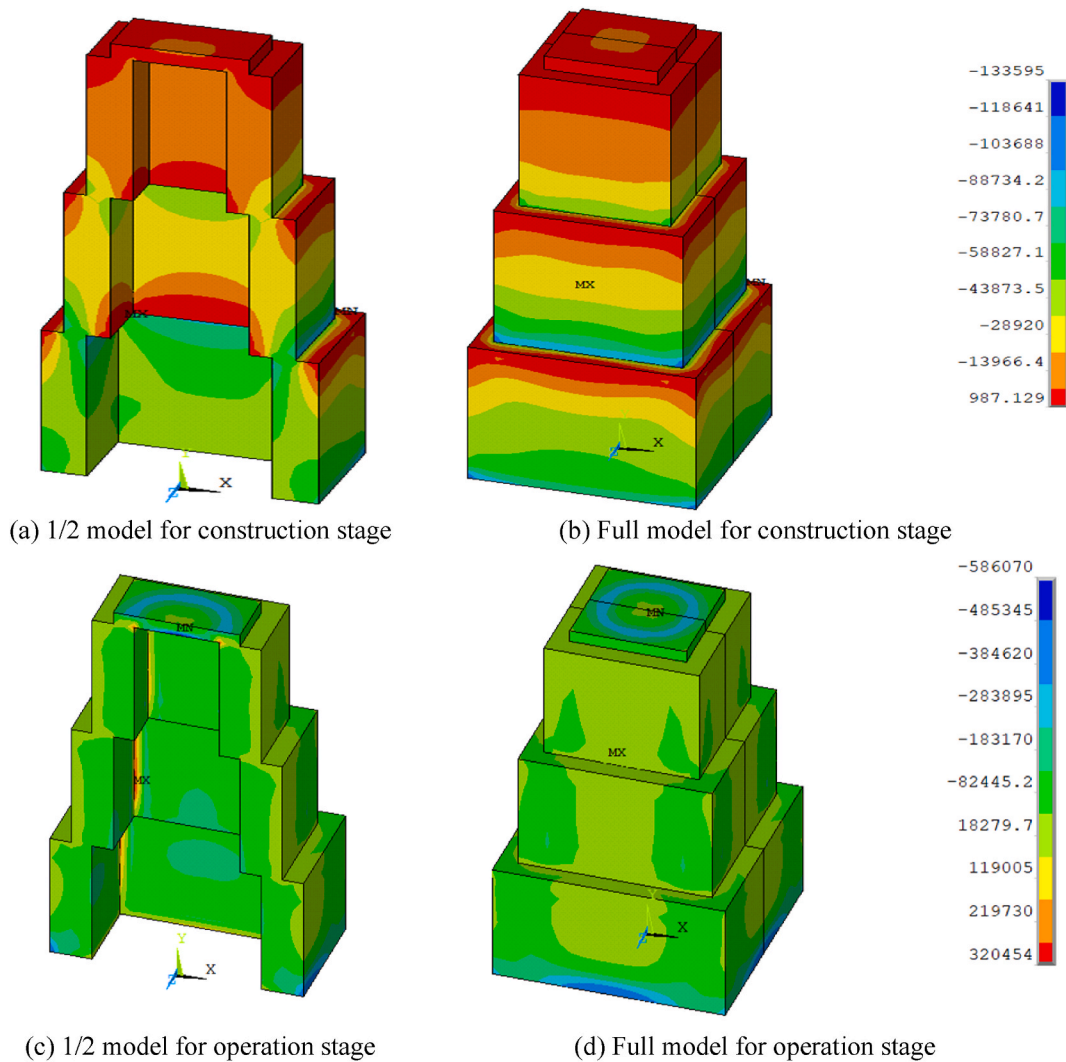


Fig. 19. Principal compressive stress of two scenarios.

- 1) Novel high-strength ISRU brick fabrication: A process combining mechanical compaction (optimal 40 MPa) with optional subsequent thermal treatment (up to 1000 °C) was developed using Martian regolith simulants. Compaction alone yielded significant compressive strengths (>20 MPa) with nano-SiO₂ at ambient temperature, while thermal treatment substantially enhanced strength (up to 44.5 MPa). This primarily physical, water-limited densification method effectively circumvents Martian environmental constraints on conventional hydration. The systematic parameter study provides essential data for adaptable ISRU material design based on performance needs and available energy.
- 2) Optimized sulfur mortar bonding: Sulfur mortar proved a robust bonding agent for the ISRU bricks. Crucially, optimized flat-cut interfaces yielded superior average bond strengths (>2.0 MPa) and shifted the failure mode from the interface to the brick material itself, indicating a highly reliable joint. This practical interface preparation offers an effective technique for structural assembly on Mars.
- 3) Modular pyramid habitat concept: A modular construction strategy was proposed utilizing the developed ISRU bricks and sulfur bonding to create pyramid-shaped habitats. This design leverages the pyramid's inherent geometric stability for resistance against anticipated Martian environmental loads, including wind, potential seismic activity, and significant internal pressure differentials.
- 4) FEA structural validation: FE analysis was employed to assess the structural integrity of the proposed pyramid habitat. Critically, the model incorporated experimentally derived non-linear stress-strain behavior representative of the ISRU bricks (e.g., ~22 MPa strength from compaction without high heat) and the characterized minimum bond strength (tensile limit ≥ 1.2 MPa) from the sulfur mortar tests. Simulation results under combined Martian gravity and operational internal pressure (1 atm differential) indicated that the maximum predicted principal tensile stresses (1.15 MPa) remained below the demonstrated capacity of the sulfur bond, validating the structural feasibility of the design even with bricks produced via the lower-energy compaction route.

CRedit authorship contribution statement

Wei Sun: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Jiabao Zhao:** Writing – original draft, Investigation, Data curation. **Handong Yan:** Writing – review & editing, Investigation. **ShaoFei Jiang:** Writing – review & editing, Validation. **Tiejiong Lou:** Writing – review & editing, Project administration, Funding acquisition.

Declaration of competing interest

All of authors declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence this work entitled **Advanced Martian construction: High-strength ISRU bricks, robust sulfur bonding, modular assembly, and FEA-verified pyramid habitats**.

Acknowledgements

The support of the internal fund at Huaqiao University [grant numbers 605-50Y22018 for Wei Sun], Xiamen Foundation for Distinguished Young Scholar [grant numbers 605-52423068 for Wei Sun], Xiamen Double Hundred Talents Program [grant numbers 605-52424055 for Wei Sun], Portuguese Foundation for Science and Technology, grants number 2022.04729.CEECIND/CP1714/CT0010, UIDB/00285/2020 and LA/P/0112/2020, Key Lab for Intelligent Infrastructure and Monitoring of Fujian Province Key Lab for Intelligent Infrastructure and Monitoring of Fujian Province, Higher-educational Engineering Research Centre for Intelligence and Automation in Construction of Fujian Province, College of Civil Engineering at Huaqiao University are gratefully acknowledged.

Data availability

Data will be made available on request.

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